

The Determinants of terrorism in West Africa: A Subnational Actor-based Analysis.

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Abstract

Following recent advancements in terrorism literature and the inclusion of spatial analysis, this paper seeks to study terrorism in West Africa from an actor-based perspective to understand the factors influencing terrorism in the region. We examine the heterogeneity of these factors for the top seven terrorist organizations operating actively in West Africa. We approximate and deploy a binomial logistic model for each terrorist organization. This methodology allows for the close examination of each group, and all terrorist groups combined to understand how economic, political, demographic, and religious characteristic of locations influence the decision-making process of terrorist groups when deciding which areas to target. Our study demonstrates that gross domestic product per capita, infant mortality rates, the share of Christians and Muslims, relative wealth index, and local government index remain the core factors influencing terrorism in the region. We find that the likelihood of attacks on cells reduces as gross domestic product per capita rises. Additionally, we find that the probability of attacks also reduces as states develop higher levels of local government control. This finding is relevant for domestic policy within individual African countries as governments could attempt to build local government capacity, while simultaneously engaging in economic development strategies.

Keywords: Terrorism, West Africa, Spatial Analysis, Heterogeneity Grid analysis, Islamic State

1. Introduction

The watch on terrorism in Africa continues as several countries, particularly in the Sahel region and the Horn of Africa, remain constantly challenged by terrorism. A comprehensive review of terrorism in West Africa reveals a refined narrative shaped by historical tensions and contemporary geopolitical influences. The turn of the 21st century witnessed a paradigm shift with the infiltration of jihadist ideologies, notably embodied by al-Qaeda in the Islamic Maghreb (AQIM) operating in the Sahel (Council on Foreign Relations, 2021). The last decade marked the ascent of Boko Haram in Nigeria, a militant group with significant implications for regional security (International Crisis Group, 2015). Recent years have also seen the emergence of the Islamic State in West Africa Province (ISWAP), also a splinter group from Boko Haram, further complicating the security landscape (Institute for Security Studies, 2019). Notably, the Global Terrorism Index for 2023 identifies four African countries among the top 10 nations most affected by terrorism (Institute for Economics & Peace, 2023). Burkina Faso ranked second, while Mali was at number four. Nigeria and Niger are also among the top 10 impacted nations at number 8 and 10, respectively.

Studying terrorism in West Africa is of paramount importance given the specific and evolving challenges faced by the region. The geopolitical significance, coupled with the profound socio-economic and political impacts of terrorism, demands a focused and comprehensive understanding of the trends. Terrorism in West Africa has far-reaching socio-economic consequences. Attacks often result in displacement, disruption of essential services, and the loss of livelihoods for affected communities. The role of state institutions in addressing terrorism cannot be overstated. A comprehensive approach involves strengthening governance, promoting the rule of law, and engaging in community-based initiatives (Jackson et al., 2011). By studying terrorism in West Africa, policy makers can better identify the weaknesses within state institutions and implement

targeted reforms to enhance their effectiveness in preventing and responding to security threats. West Africa's complex landscape of terrorism involves the intersection of local insurgencies with global jihadist networks. The study of these connections is essential for recognizing and disrupting the transnational nature of extremist activities. It enables a more nuanced understanding of how international dynamics contribute to regional security challenges, emphasizing the need for coordinated efforts at both local and global levels (Le Sage, 2007).

The application and advancements of spatial analysis in terrorism literature have yielded valuable insights, incorporating the geographical attributes of locations into studies (Python, Andreas, et al., 2021; Python, Janine, et al., 2017; Walther et al., 2020). However, it focuses on aggregated terror incidences with no distinction to different terrorist organizations involved and ineluctably ignores the heterogeneity in the determinants of terrorism. One exception is Metiki (2021), that groups terrorist activities based on the ideology of the groups involved. However, his analysis does not include a spatial domain. Consequentially, there is a need to explore the diverse determinants of terrorist attacks specific to each organization.

In this study, we propose a framework to elucidate the occurrence of terrorism perpetrated by the seven most prominent terrorist organizations in West Africa. Our analysis draws on data from the Armed Conflict Location and Event Data Project (ACLED) database for the period between 2003 and 2022, covering six West African countries. We include spatial data on economic, political, religious, demographic, and geographic variables, which may influence choice of target locations for terrorist groups. We extract all variables into a 0.5° spatial grid with R and create dummy variables for each terrorist organization and one dummy for all groups together. These variables are coded as '1' if that group attacked a cell and '0' if not. We fit the data on multiple

independent binomial logistic regressions for each terrorist organization, which enables us to pinpoint and compare the influence of different terrorism drivers across various groups. The first model excludes variables of religious parity, while the second model includes them. The Akaike Information Criterion shows that all regression models with these religious variables explain more variation in the data. The findings highlight the significance of relative wealth, gross domestic product (GDP) per capita, population density, proximity to national borders and share of Christians and Muslims as factors influencing terrorist attacks in West Africa. Particularly, cells with higher GDP per capita have a lesser risk of terrorist attacks, confirming that poverty is a focus of terrorism in West African countries. An exception is the JNIM, which is more likely to attack with a rise in cell GDP per capita. The odds that any group attacks reduce as local government control increases. Furthermore, the likelihood of attacks reduces for cells farther from the border. Except for JNIM, Communal militia and Fulani ethnic militias, which are more likely to attack cells farther from a national frontier.

1.2 Statement of Problem

In recent years, West Africa has experienced a distressing surge in terrorism, resulting in a significant loss of life and property. The Sahel region in sub-Saharan Africa has emerged as the epicentre of terrorism, surpassing both South Asia and the Middle East and North Africa (MENA) combined in terms of terrorism-related fatalities in 2022. In 2022, deaths in the Sahel accounted for 43 percent of the global total, a stark increase from just one percent in 2007 (Institute for Economics & Peace, 2023). These figures underscore the urgent need for a comprehensive examination of the factors influencing the occurrence of terrorism in West Africa. Terrorism poses a threat to lives and properties, prevents economic growth and foreign investment, significantly impacts the

tourism industry and causes long-term economic stagnation. The core problem addressed by this research is to unravel the heterogeneity of factors contributing to the decision-making processes of terrorist groups operating in West Africa. By closely examining economic, political, demographic, and religious variables at the spatial level, this paper seeks to identify key determinants shaping the selection of target areas by these terrorist organizations. Also, this study aims to contribute to the existing literature on terrorism through a better understanding of the patterns of attacks from different terrorist groups. Specifically, this study aims to: determine the factors influencing the choice of target locations for the top seven terrorist organizations in West Africa.

2. Literature Review

2.1 Conceptual Review

2.1.1 Concept of Terrorism

Terrorism encompasses acts of violence carried out by non-state organizations against representatives of a nation-state. It involves politically, religiously, or ethnically motivated attacks outside legitimate warfare perpetrated by non-state organizations targeting representatives of a nation-state to communicate to a large audience to achieve political, religious, or other goals. Adu and Osadola (2022) characterise terrorism as the illegal utilization of force or violence against individuals or property, with the intention of intimidating or coercing a government, civilian population, or any subset thereof, in pursuit of political or social goals. Terrorism involves one violence-producing actor (for example, the Islamic State) and another actor who may or may not produce violence. For example, civilians attacked in an altercation cannot retaliate, but the military or police can attempt to contain the situation to the extent of their ability. It is important to note that these attacked actors must be representatives of a nation-state where the intended

message passes through to. Specifically, representatives could range from civilians, military, and church buildings to state facilities. Data from ACLED is filtered to include only events that match the above definition of terrorism.

2.1.2 Terrorism in West Africa

Over the past few years, West Africa has witnessed a concerning evolution in the landscape of terrorism (Blanchard, 2021). In recent years, the region has experienced an upsurge in terrorist activities, with several groups gaining prominence. The multifaceted factors contributing to this phenomenon include economic grievances, political instability, and religious extremism (United Nations Office for West Africa and the Sahel, 2020).

In Burkina Faso, Islamist extremist groups associated with both al-Qaeda and the Islamic State, have intensified their attacks, targeting civilians, security forces, and governmental institutions. The northern regions, in particular, have become hotspots of violence, with an alarming increase in displacement and humanitarian crises (International Crisis Group, 2020). Mali, situated in the Sahel region, has faced persistent violence from groups like JNIM. The central and northern regions, including areas such as Mopti and Timbuktu, have been focal points of insurgent activities. The complex ethnic and political landscape further complicates counterterrorism efforts, posing challenges to establishing lasting stability (BBC News, 2021). Benin, although not at the epicentre of terrorism, has faced challenges in maintaining security due to its proximity to high-risk areas especially in the northern part. In Nigeria, Boko Haram has continued to be a major player in the region, showcasing adaptability in tactics and strategies (Campbell, 2019). The northeastern regions of Nigeria, particularly Borno state, have borne the brunt of the insurgency. Additionally, the expansion of affiliated groups such

as the Islamic State in West Africa Province (ISWAP) has further escalated the security crisis. The Sahel region, encompassing countries like Mali, Burkina Faso, and Niger, has become a hotspot for terrorist activities, posing a significant threat to regional stability.

Table 1. Distribution of Event types and top 7 Terrorist groups.

	All events	Benin	Burkina Faso	Cameroon	Chad	Mali	Niger	Nigeria	Togo
	%	%	%	%	%	%	%	%	%
Event type									
Battles	20,4	46,15	15,51	28,86	19,47	13,88	41,18	19,4	37,5
Explosions/Remote violence	8,92	10,26	7,36	7,18	15,52	7,21	20,92	8,53	25
Strategic developments	14,52	15,38	32,11	15,81	15,31	22,08	2,61	2,81	.
Violence against civilians	56,16	28,21	45,02	48,14	49,7	56,83	35,29	69,26	37,5
Damage									
No fatalities	48,66	51,28	69,96	53,36	52,25	59,64	31,37	31,24	31,25
At least one	51,34	48,72	30,04	46,64	46,75	40,36	68,63	68,76	68,75
Yearly interval									
2003-2007	0,14	.	.	.	.	.	.	0,37	.
2008-2012	3,21	.	.	.	.	.	.	8,63	.
2013-2017	12,63	.	0,23	18,05	4,47	5,99	20,55	20,89	.
2018-2022	84,03	100	99,77	81,95	95,53	94,01	79,45	70,11	100
Top 7 terror groups									
Ambazonian Separatists	7,83	.	.	99,84	.	.	.	0.16	.
Boko Haram	8,02	.	.	13,56	.	11,29	0,78	74,37	.
Communal Militia	11,77	.	.	.	.	.	.	100	.
Fulani Militia	10,57	.	.	0,06	9,63	.	.	90,31	.
Islamic State	34,75	0,11	10,5	33,25	6,91	17,25	2,59	29,4	.
JNIM	23,84	0,84	63,82	.	29,12	5,8	.	.	0,42
Katiba Macina	3,22	0,19	36,84	.	62,96	.	.	.	.
Total no incidents	15916	39	3192	3257	153	1972	1318	5969	16
Share in total incidents	100	0.25	20,05	20,46	0,96	12,39	8,28	37,5	0,1

Source Author's compilation

Terrorist incidents in West Africa date back to the early 2000's, with a gradual rise starting in 2010 and an increment in the frequency and severity of attacks from 2014 onward. Table 1 shows that incidences from 2003 - 2007 made up 0.14%, from 2008 - 2012 made up 3,21%, from 2013 - 2017 made up 12,63%, and from 2018 - 2022 made

up 84,03% of total incidence which stood at 15 916. Violence against civilians remains a top activity for terrorists, especially in Nigeria, where it makes up 69% of all incidences recorded in the region. Notably, 51% of all incidences recorded at least one fatality. Table 1 depicts the top seven terrorist organizations present in West Africa. The JNIM, an Islamist militant organization with missions to establish a strict form of Islamic law and governance in the areas it operates, represent (23,84%) of all incidences, and is present in Burkina Faso, Chad, Mali, and parts of Benin. The Islamic State, a jihadist extremist group with a mission to establish an Islamic caliphate in West Africa, represents (34,75%) of all incidences, and is operative in Nigeria, Niger, Chad, and Cameroon. Boko Haram, established to enforce a strict form of Islamic law in the region and to create an Islamic state governed by their interpretation of Sharia law, represents (8,02%), and is present in Cameroon, Mali, and Niger while dominating in Nigeria. The Fulani ethnic militia, present in Nigeria and Chad, make up 10,57% of all incidences recorded. Leading in insurgences, Nigeria records the most terrorist incidences in the region at 37,5% of all recorded occurrences. Cameroon, Burkina Faso and Mali follow behind with 20,46%, 20,05%, and 12,39%, respectively, of all recorded occurrences.

Figure 1 shows the yearly evolution of terrorist activities in West Africa from 2003 to 2022. It depicts those early occurrences of terrorism started in the North-Eastern part of Nigeria in the late 2000's and began spreading across the region in 2014 and worsens overtime. The figure shows the spread of terrorist activities from Nigeria to the borders of Cameroon and Chad, and later from 2014 onwards, to Mali, Burkina Faso and Niger.

The division between 'domestic' and 'international' terrorism corresponds to cases where members of the organization target representatives of a state with which they

share a territorial or political affiliation, or they attack representatives of foreign states (Enders and Sandler, 2002).

Figure 1. Yearly Evolution of Terrorist Activities in West Africa



Source Author's compilation

2. 2 Empirical Review

The watch on terrorism in Africa continues as several countries, particularly in the Sahel region and the Horn of Africa, remain constantly challenged by terrorism. Since the wake of the 7/11 attack on the United States of America, the international community has taken several actions against the rise of world terror. In the first 100 days after the drastic event that shook the hearts of many and took the lives of many, the United States began to destroy al-Qaeda's grip on Afghanistan by driving the Taliban from power, disrupting its global operations and financing networks (United States Department of States, 2001). The USA began a worldwide hunt for terrorists with the help of allied nations. They launched a military campaign to suppress any ongoing terrorist operation. These ongoing complications led to various coalitions, changes in territory, and recruitment for many terrorist organizations in more conducive areas like Africa. Falode (2016) posits that

passage of leadership in terrorist organizations can often lead to changes in objectives, methods and general reasoning.

Terrorist activities have persisted for over eight years in the Sahel region, where JNIM and the Islamic State in the Greater Sahara (ISGS) are fuelling insurgencies in Mali, Burkina Faso, and Niger. Since ISGS emerged in 2015, these two rival jihadi groups coexisted in an uneasy alliance for several years. In 2020, the United Nations monitoring team tracking the global jihadi threat alerted the Security Council to the ongoing operational success of Islamic State factions in West Africa and the Sahel, as well as those of al-Qaida, (Héni and Caleb, 2020). Hamidu (2022) suggests that terrorist organizations could also exploit border insecurity present in the region as opportunities for recruitment, training, and radicalization of youth, both within and across borders, enabling the execution of their plans against enemies or targets. Adu and Osadola (2022) argue that although identity politics plays a trivial role in the survival and continued existence of sovereign African states, the ideology of good neighbourliness fuelled by identity politics has led to the spread of insurgence across international borders and heightened cooperation between groups within the region.

The factors affecting terrorism are multifaceted and may include economic drivers like national income levels. Particularly, Metiki (2021) finds that terrorist attacks increase as we move from countries with high-income quartiles to countries with low-income quartiles and confirms that poverty is a breathing ground for terrorism. Daniel and Thomas (2013) find that the relationship existing between terrorism and economic growth is heterogeneous over time and across space. They argue that different causal mechanics across countries associated with factors specific to each country, including the degree of development in politics and economics, cultural ties, the severity and continuity of

terrorist actions, and political instability, govern its resilience to terrorism's impact and the degree to which economic motives drive terrorist activities.

The link between terrorism and economic growth is quite complex, and it varies with factors like development, culture, and regional instability, often driven by dissatisfaction and economic disparities, especially in Africa, where social conditions and ethnic or religious tensions contribute to extremism. Graham, Brock, and Gregory (2008) conclude that terrorism could serve as a method to voice discontent with the current state of affairs. Economically speaking, this implies that the frequency of terrorist acts might correlate with the level of income inequality. Midlarsky (2011) discusses how societal conditions like inequalities, social grievances, and ethnic or religious tensions fuel the rise of political extremism and its escalation into mass violence. Terrorism motivated by political ideologies in sub-Saharan African countries is low, whilst most groups in the region strive for the dominance of their ethnicity or the supremacy of their religion (Metiki, 2021).

Spatial analysis functions as a geospatial compass, guiding economists and data analysts through the intricate spatial domain of data. It lies at the intersection of geography, data science, and social sciences and plays a pivotal role in our understanding of complex spatial phenomena. Additionally, spatial analysis harnesses the power of geospatial data and advanced computational techniques to unearth insights often hidden in the spatial patterns of our world. While the influence of geography on human agency is a subject of debate in scholarly discussions, the inclusion of location-based factors in these models acknowledges the inherent geographic variations in resources for political action. It also recognizes that some essential influences on human agency can be identified and quantified in the specific geographic contexts where violent events occur (Walther et al., 2020). Studies of social conflict using georeferenced conflict data have

shown that environmental factors like rainfall levels and weather conditions heighten the likelihood of conflict at the local level. (Josiah et al., 2018). Their argument revolves around the influence of accessibility on the probability of a terror attack, suggesting that targets situated close to international borders or within urban areas face a higher risk of being targeted. Both jihadist groups and criminal gangs prosper in areas lacking government control, often found in rural regions far from capital cities and close to national borders. The primary Islamist organizations, ISGS and JNIM, seek recruits and resources in these lawless areas, and they sometimes establish agreed-upon locations of influence, even engaging in fierce competition for the same territory (Orosz, 2022).

Choosing the appropriate grid size is crucial for effectively analysing political violence across a vast region. Following existing literature, a 50 km by 50 km grid strikes a balance, providing enough events in each cell for meaningful analysis while offering a localised view of political violence. This approach facilitates comparisons of conflict evolution across a broad region, enabling a more consistent representation of violence compared to varying administrative units that differ in size significantly, especially in sparsely populated areas like the Sahara (Walther et al., 2020). Python, Andreas, et al. (2021) use regular cells at 0.5° spatial resolution to explain the occurrence of terrorism worldwide, Walther et al. (2020) use 50km by 50km grid cells to present the spatial conflict indicator that measures conflict intensity and the spatial distribution of conflict within a region. Tollefsen et al. (2015) organize and maintain the PRIO-GRID dataset, a standardized spatial grid structure with global coverage at a resolution of 0.5° decimal degrees. Recent literature on terrorism incorporating a spatial domain supports the usage of grid cells sized at 55 km by 55 km.

3. Methods and Materials

3.1 Terrorism database

We gathered the data used in this study from secondary sources. The data on terrorism comes from the Armed Conflict Location and Event Data Project (Raleigh et al., 2010). Detailed information on the names of terrorist groups involved in any attack and the type of attack is well specified under the columns ‘actor1’ and ‘event-type’, respectively. They also include fatality counts and the geo-locations information in the latitude and longitude columns. We filtered the data to match the definition of terrorism employed in this paper. We extracted the groups responsible for attacks in the study area from 2003 to 2022, the associated fatalities, and geo-locations. It represents a total of 15 916 events for six countries. We dropped Benin and Togo because they recorded few incidences of terrorism. We also dropped unidentified groups and all terrorist groups that didn’t meet our threshold of 500 total attacks from 2003 to 2022, reducing the number of active terrorist groups to 7 and the number of incidences to 15861.

3.2 Covariates

3.2.1 Economic covariates

We consider the role of the following economic factors:

- Gross Domestic Product (GDP) per capita, compiled from Jiandong and Gao (2021), global 1 km by 1 km gridded gross domestic product database. Metiki (2021) posits that poverty is a focus of terrorism. We expect that the probability of an attack from any group should reduce with higher GDP per capita.
- Electricity consumption which measures economic activity and is computed from night light data (Jiandong and Gao, 2021). We expect that the probability of an attack from any group should increase for low cell electricity consumption.
- Relative wealth index which is compiled from the Humanitarian Data Exchange (Chi et al., 2022), and measures the standard of living (relative) within countries.

3.2.2 Political covariates

We include the following covariates to measure the level of democracy in countries. Each of the selected variables range from 0 to 1 with ‘1’ meaning the highest possible level has been attained and ‘0’ no level attained. The dataset is a country-year datasets, so we extracted the following variables from 2003-2022:

- Electoral democracy index: it measures the degree to which a country achieves the ideal of electoral democracy comprehensively (Coppedge et al., 2023).
- Liberal democracy index: it underscores the significance of safeguarding individual and minority rights from both state oppression and the dominance of the majority (Coppedge et al., 2023).
- Deliberative democracy index: it centres on the decision-making mechanisms within a political system. Deliberative processes ensue when public discourse prioritizes the collective welfare and drives political choices (Coppedge et al., 2023).
- Egalitarian democracy index: Egalitarian democracy is attained when the rights and freedoms of individuals are uniformly safeguarded across all social strata, and resources are equitably distributed among all societal groups (Coppedge et al., 2023).
- Equality before the law and individual liberty index: measures the degree to which public administration upholds legislation transparently and impartially, and the extent to which citizens enjoy access to justice, secure property rights, freedom from forced labour, freedom of movement, physical integrity rights, and freedom of religion (Coppedge et al., 2023).
- Local government index measures the degree to which elected local government officials can function autonomously without interference from unelected entities at the local level (Coppedge et al., 2023).

3.2.3 Religious covariates

We consider the role of religious division, proxied by the percentage of Muslims and Christians in each cell. We use Carl and Hunziker (2018) spatially interpolated data on ethnicity. Python, Janine, et al. (2017) suggest that for ethnically polarized societies, terrorists may intentionally target specific ethnic groups to exacerbate tensions, potentially escalating into ethnic conflict. We assume the same for religious cases.

3.2.4 Demographic covariates

We consider the role of the following demographic factors:

- We include data on infant mortality from Centre for International Earth Science Information Network (2021) high-resolution data on infant mortality. We relate poor economic conditions to high levels of infant mortality, so we expect that the probability of an attack from any group should increase with higher levels of infant mortality rate.
- We include data on population density from Gridded Population of the World V4 (Centre for International Earth Science Information Network, 2018). They provide data on a five-year basis from 2000 to 2020. We expect the probability of an attack from any group should be higher in densely populated areas.
- We include data on road network density from GLOBIO GRIP dataset (Meijer et al., 2018). We assume that the ease of movement in various areas may facilitate terrorist activities.

3.2.5 Geographic covariates

We consider the influence of geographical factors such as altitude and proximity to the nearest national border. Nemeth, Mauslein, and Stapley (2014) suggest that the distance to a national frontier and elevation might impact terror incidences. We extract altitude data from NOAA National Geophysical Data Centre (2009) and NOAA Technical Memorandum: Amante and Eakins (2009).

3.3 Data Preparation

We filtered the ACLED data to include only events where the primary actor (actor 1) was a terrorist group and the actor on the receiving end (actor 2) were civilians. We exclude events where two groups clash or the police attacking terror groups. Subsequently, we excluded all terrorist groups that whose number of incidents did not sum up 500. For the spatial domain, we generated a grid of regular cells from the study area and obtained 1950 equally sized square grid cells of 55.5 km x 55.5 km. We spatially joined the terrorism dataset and the grid together and then aggregated by year and grid ID. The resulting dataset was a long panel data containing, for each cell and each year, the terrorist groups

that attacked, the number of times in that year which the group struck and the total fatality from that group within that year. We extracted all variables into the grid. Subsequently, we merged the two datasets by grid ID and year. The output is a long panel data frame containing the grid ID, year, terrorist group, count, fatality and 16 explanatory variables.

Table 2. Descriptive Statistics

Variables	Min.	1 st Qu.	Median	Mean	3 rd Qu.	Max.	NA's
GDP	0.000588	0.008997	0.012	0.137	0.069	27.725	3
Electricity consumption	0	0	1356	5799	4093	939686	3
Road density	0	6.167	51.361	64.733	96.389	1480.97	522
Infant mortality	0	37.12	61.47	55.25	72.36	134.52	501
Population density	0.028	0.361	7.851	48.214	40.375	7575.438	
Relative Wealth Index	-1.094	-0.528	-0.456	-0.431	-0.369	0.865	13601
Electoral democracy index	0.14	0.28	0.49	0.44	0.566	0.687	
Liberal democracy index	0.031	0.12	0.32	0.27	0.4	0.5	
Deliberative democracy index	0.1	0.135	0.362	0.331	0.465	0.574	
Egalitarian democracy index	0.069	0.183	0.294	0.274	0.386	0.456	
Equality before the law and individual liberty index	0.166	0.501	0.672	0.614	0.764	0.848	
Local government index	0.039	0.367	0.702	0.625	0.879	0.968	
Elevation	-236.5	289.4	363.3	422.7	491.8	1962.9	
Distance	0.053	35.445	100.541	124.243	187.949	455.101	
Share of Christians	0	0.006	0.02	0.088	0.053	0.848	
Share of Muslims	0	0.05	0.96	0.648	0.983	0.994	

Source Author's compilation

3. 4 Outcome Variable

We have eight outcome variables. The first outcome variable represents terror attack which is coded as '1' if the cell had any terrorist attack in that particular year and '0' if there was no attack. The seven left represent each terrorist group and each is coded as '1' if a cell was attacked by that organization and '0' if there was no attack. We created dummy variables for each terrorist group and one variable representing all groups, where:

$$group_{variable} = \begin{cases} 1, & \text{if group attacked in cell} \\ 0, & \text{Otherwise} \end{cases} \quad (2.1)$$

The nature of the dependent variable is binary, with $y_i \in [0, 1]$. An appropriate approach is the binomial logistic model, a statistical tool employed to examine the correlation between one or more independent variables and a binary dependent variable. We wish to estimate the unknown probability that an individual cell i is attacked by any group given a set of independent variables $X_1, \dots, X_n$:

$$p(x) = P(\{Y = 1\} | \{X_1, \dots, X_n\}) \text{ with } x = (x_1, \dots, x_p), \quad (2.2)$$

The logit transformation function.

$$\text{logit}(y) = \ln\left(\frac{y}{1-y}\right) \in \mathbb{R}, y \in]0, 1[\quad (2.3)$$

Its inverse is the function.

$$\text{logit}^{-1}(y) = \frac{\exp(y)}{1 + \exp(y)} \in]0, 1[, \quad y \in \mathbb{R} \quad (2.4)$$

The binary logistic regression model is characterised by:

$$\forall i \in 1, \dots, n$$

- $(x_1, i, \dots, x_p, i)$ is a realization of the random real vector $(X_1, \dots, X_p)$;
- Knowing $(X_1, \dots, X_n) = (x_1, \dots, x_p, i)$, $x_i y_i$ is a realisation of:

$$Y_i \sim \mathcal{B}(p(x_i)), \quad p(x_i) = \mathbb{P}(\{Y = 1\} | \{(X_1, \dots, X_p) = x_i\}), \quad (2.5)$$

- We have:

$$\text{logit}(p(x)) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p, \quad x = (x_1, \dots, x_p), \quad (2.6)$$

Where $\beta_0, \dots, \beta_p$ are unknown real coefficients.

Thus, $p(x)$ and $\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$ are linked by a logit transformation called the logit link.

We deduce the expression of $p(x)$:

$$p(x) = \text{logit}^{-1}(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p) = \frac{\exp(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)}{1 + \exp(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)} \quad (2.7)$$

We estimate $\beta_0, \dots, \beta_p$ from data by calculating $p(x)$ by substitution. To do that, the model uses the maximum likelihood method.

4. Results and Discussion

4.1 Baseline results

In this section, we present the very first estimation results. Here, we exclude the variables representing religious division from the model. In Table 3 and Table 5, we have grouped the estimation results of the variables often used in the literature on terrorism. Each column shows the results for one of the seven identified terrorist organizations, with the first column for all groups combined. All models are estimated using a binomial logistic regression. They include sets of year dummies and region dummies. Incorporating fixed effects for year and country in our model is crucial, particularly because we are working with panel data that exhibits variation across time and among different countries.

Table 3 presents estimation results for the Binomial Logistic model. The analysis of Table 3 provides a first key piece of information: GDP per capita is significant at the 10% level. This means that GDP per capita is correlated with terror attacks in West Africa. The negative sign on the coefficient (-0,101) indicates that cells with higher GDP per capita have lesser risks of terrorist attacks, confirming that low national income can be associated to terrorist activities in West African countries. With an exception to JNIM (significant at the 5% level), whose chances of attack increases at places with higher GDP per capita. Population density is also significant at 10% level of significance for all identified, but at 1% for the Ambazonian Separatists and Katiba Macina. The likelihood of terrorist attacks reduces for densely populated cells. However, these chances increase for the Katiba Macina, Fulani, and Communal militia individually, who are more likely to attack densely populated areas.

Table 3. Binomial logistic model estimation results in Log Odds.

Variables	All identified	Ambazonian Separatists	Boko Haram	Communal Militia	Fulani Ethnic Militia	Islamic State	JNIM	Katiba Macina
GDP per capita	-0,101*** (0,034)	0,454 (0,485)	-0,005 (0,143)	-0,018 (0,220)	-0,013 (0,103)	-0,056 (0,167)	15,906** (7,536)	-5,870 (5,974)
Population density	-0,001*** (0,0002)	-0,009* (0,005)	-0,004*** (0,001)	0,003*** (0,001)	0,005*** (0,001)	-0,004*** (0,001)	0,0002 (0,003)	0,008* (0,004)
Democracy index	0,202 (0,451)	-24,559 (25,097)	-4,532*** (1,602)	13,013** (5,541)	9,062*** (1,535)	-2,677*** (0,818)	1,123 (1,004)	5,276** (2,385)
Local government index	-0,856*** (0,207)	2,530 (5,725)	0,864 (0,562)	1,142 (1,139)	0,543 (0,580)	-0,117 (0,359)	-3,115*** (0,992)	11,330*** (3,133)
Distance to border	-0,002*** (0,0003)	0,003 (0,007)	-0,00001*** (0,00000)	0,004*** (0,001)	0,005*** (0,001)	-0,005 (0,001)	0,009*** (0,002)	-0,008*** (0,003)
Elevation	0,0001 (0,0002)	0,002** (0,001)	0,001*** (0,0005)	0,002*** (0,001)	-0,002*** (0,0004)	-0,0003 (0,0004)	-0,007*** (0,002)	0,009*** (0,003)
Infant mortality	-0,002** (0,001)	-0,039*** (0,012)	0,001 (0,001)	0,004 (0,004)	-0,012*** (0,002)	0,017*** (0,002)	0,002 (0,003)	-0,011** (0,004)
Road density	-0,0005 (0,0003)	0,003 (0,003)	0,005 (0,001)	-0,003 (0,002)	-0,010*** (0,001)	0,003*** (0,001)	0,002 (0,002)	-0,003 (0,003)
Rel. Wealth Index	2,428*** (0,232)	3,571 (3,151)	3,519*** (0,833)	-7,995*** (1,066)	-0,381 (0,635)	0,702 (0,565)	-0,018 (0,981)	1,503 (1,533)
AUC	0,5	0,94	0,75	0,69	0,76	0,69	0,85	0,59
No. observations	25 233	2 487	2 487	2 487	2 487	2 487	2 487	2 487
No. attacks	15 916	1 246	1 276	1 873	1 682	5 531	3 795	513
Year fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source Author's compilation

All models are estimated using a binomial logistic regression. They include sets of year dummies and region dummies. Estimation results are presented in log odds, with standard errors in parentheses. For the sake of readability, statistically significant coefficients are with bold stars. *p<0,1; **p<0,05; ***p<0,01

Democracy's impact on the likelihood of attack varies among group. The democracy index is insignificant for all aggregate groups but gains significance for groups individually. The index is significant at the 5% level for Katiba Macina and Communal militia but significant at the 10% level for Boko Haram, Fulani Militia and Islamic State. The chances of Boko Haram and Islamic State attacking reduce for higher levels of democracy. Also, the Communal militia, Fulani ethnic militia, and Katiba Macina increase the chances of attacks for cells with a higher democracy index.

Table 4. Binomial logistic model estimation results (Odds Ratio)

Variables	All identified	Ambazonian Separatists	Boko Haram	Communal Militia	Fulani Ethnic Militia	Islamic State	JNIM	Katiba Macina
GDP per capita	0.904*** (0.034)	1.575 (0.485)	0.995 (0.143)	0.983 (0.220)	0.987 (0.103)	0.945 (0.167)	8,089,988** (7.536)	0.003 (5.974)
population density	0.999*** (0.0002)	0.991* (0.005)	0.996*** (0.001)	1.003*** (0.001)	1.005*** (0.001)	0.996*** (0.001)	1.000 (0.003)	1.008* (0.004)
Democracy index	1.223 (0.451)	0.000 (25.097)	0.011*** (1.602)	448,388.7** (5.541)	8,619.729*** (1.535)	0.069*** (0.818)	3.073 (1.004)	195.551** (2.385)
Local government index	0.425*** (0.207)	12.557 (5.725)	2.101 (0.562)	3.132 (1.139)	1.721 (0.580)	0.890 (0.359)	0.044*** (0.992)	83,293.210*** (3.133)
Distance to border	0.998*** (0.0003)	1.003 (0.007)	0.988*** (0.00000)	1.004*** (0.001)	1.005*** (0.001)	0.995 (0.001)	1.009*** (0.002)	0.992*** (0.003)
Elevation	1.000 (0.0002)	1.002** (0.001)	1.001*** (0.0005)	1.002*** (0.001)	0.998*** (0.0004)	1.000 (0.0004)	0.993*** (0.002)	1.009*** (0.003)
Infant mortality	0.998** (0.001)	0.962*** (0.012)	1.011 (0.001)	1.004 (0.004)	0.988*** (0.002)	1.017*** (0.002)	1.002 (0.003)	0.989** (0.004)
Road density	1.000 (0.0003)	1.003 (0.003)	1.005 (0.001)	0.997 (0.002)	0.990*** (0.001)	1.003*** (0.001)	1.002 (0.002)	0.997 (0.003)
Rel. Wealth Index	11.337*** (0.232)	35.541 (3.151)	39.811*** (0.833)	0.0003*** (1.066)	0.683 (0.635)	2.018 (0.565)	0.982 (0.981)	4.497 (1.533)
AUC	0.5	0.94	0.75	0.69	0.76	0.69	0.85	0.59
No. observations	25 233	2 487	2 487	2 487	2 487	2 487	2 487	2 487
No. attacks	15 916	1 246	1 276	1 873	1 682	5 531	3 795	513
Year fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source Author's compilation

All models are estimated using a binomial logistic regression. They include sets of year dummies and region dummies. Estimation results are presented in Odds Ratio, with standard errors in parentheses. For the sake of readability, statistically significant coefficients are with bold stars. * $p < 0,1$; ** $p < 0,05$; *** $p < 0,01$

Proximity to national borders increases the likelihood of terrorist attacks, with exceptions for certain groups. From Table 4, we confirm that areas closer to a national border have higher chances of becoming targets. Notably, the odds of a group targeting cells decreases by 0,2% for an additional kilometre away from a national frontier. However, the opposite occurs for communal militias, Fulani ethnic militia and the JNIM, who are more likely to attack cells farther from the border. Results also show that the likelihood of attacks reduces as local government control increases. The relative wealth

index measures the relative standard of living of cells compared to other surrounding cells. The odds that a cell is attacked increases by 11,337 times for each additional unit of the relative wealth. Analysis shows that discontent and dissatisfaction with economic conditions could also increase the likelihood of terrorist attacks.

Boko Haram is less likely to attack densely populated cells and cells with high democracy index. Areas farther from the national border are less likely to be targeted as well. Results show that the likelihood of attacks from Boko Haram increases for cells with high infant mortality. Confirming that they are more likely to attack areas with poor economic conditions. The Islamic State reduces odds of targeting cells by 93%, with every additional unit of the democracy index. Suggesting that the Islamic State targets areas where democracy isn't popular. They also target areas closer to a national border as the odds reduce by 0,5% for an additional kilometre farther from the national frontier. The JNIM tends to avoid targeting regions with strong local government presence and exhibits a preference for attacking areas situated farther away from national borders.

4.1.2 Results for Binomial logistic model with religious covariates

Table 5 shows the results for the binomial logistic model with two additional variables added to the regression. These variables represent cell share of Christians and Muslims and may help us understand the religious dynamics of terrorism in the region.

The religious composition of a cell significantly impacts the likelihood of terror attacks, with the odds increasing for cells with higher proportions of Christians and Muslims. The share of Christians and Muslims is significant to all identified groups together. Other variables are not highly affected after including these two variables to the model, although, GDP per capita is no longer significant in the model.

Table 5. Binomial logistic model with religious covariates estimation results

Variables	All identified	Ambazonian Separatists	Boko Haram	Communal Militia	Fulani Ethnic Militia	Islamic State	JNIM	Katiba Macina
GDP per capita	1.003 (0.033)	0.910 (2.112)	1.139 (0.170)	1.163 (0.317)	0.831 (0.131)	1.335* (0.160)	48,882.7 (8.803)	0.00002 (6.865)
population density	0.999*** (0.0002)	0.987* (0.007)	0.998*** (0.001)	1.005*** (0.001)	1.003*** (0.001)	0.997*** (0.001)	1.000 (0.003)	1.007* (0.004)
Democracy index	0.813 (0.449)	0.000 (49.714)	0.027** (1.600)	601,412.100** (5.453)	761.732*** (1.635)	0.156** (0.900)	1.542 (1.064)	12.206 (2.446)
Local government index	0.613** (0.209)	307,887.100 (10.062)	1.583 (0.580)	3.509 (1.134)	2.459 (0.671)	0.496* (0.396)	0.069*** (1.014)	56,561.410*** (3.026)
Distance to border	0.998*** (0.0004)	0.988 (0.013)	0.991*** (0.002)	1.007*** (0.001)	1.002*** (0.001)	0.997*** (0.001)	1.010*** (0.002)	0.992*** (0.003)
Elevation	1.000* (0.0002)	1.005** (0.002)	1.001 (0.001)	1.002*** (0.001)	0.999*** (0.0004)	1.000 (0.001)	0.993*** (0.002)	1.009*** (0.003)
Infant mortality	0.998* (0.001)	0.991 (0.021)	1.003 (0.005)	0.001 (0.004)	0.992*** (0.003)	1.011*** (0.002)	1.002 (0.003)	0.992* (0.004)
Road density	0.999** (0.0003)	1.012 (0.007)	1.002 (0.001)	0.992*** (0.002)	0.997** (0.001)	1.000 (0.001)	1.002 (0.002)	0.998 (0.003)
Rel. Wealth Index	17.381*** (0.240)	876.349 (5.531)	209.982*** (0.908)	0.004*** (1.072)	0.037*** (0.806)	4.571** (0.600)	1.255 (0.983)	10.730 (1.583)
Share of Christians	10.182*** (0.228)	772.456* (4.010)	0.269 (1.020)	0.703 (1.416)	4.937** (0.730)	0.014*** (0.706)	3.742* (0.752)	2.978*** (0.959)
Share of Muslims	6.767*** (0.158)	0.0003*** (1.909)	47.360*** (0.704)	137.789*** (0.818)	0.003*** (0.541)	42.799*** (0.591)	0.679 (0.979)	0.235 (1.056)
AUC	0.59	0.98	0.77	0.76	0.81	0.73	0.84	0.6
No. observations	25 233	2 487	2 487	2 487	2 487	2 487	2 487	2 487
No. attacks	15 916	1 246	1 276	1 873	1 682	5 531	3 795	513
Year fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Source Author's compilation

All models are estimated using a binomial logistic regression. They include sets of year dummies and region dummies. Estimation results are presented in Odds Ratio, with standard errors in parentheses. For the sake of readability, statistically significant coefficients are with bold stars. * $p < 0,1$; ** $p < 0,05$; *** $p < 0,01$

Table 5 shows that the likelihood of attacks on cells increases as the share of Christians and Muslims increases. More specifically, the odds that a cell is attacked increases by 10 times, for a unit increase to the proportion of Christians in that cell. Also,

the odds that a cell is attacked increases by 6 times, for an extra unit to the share of Muslims in that cell.

The Islamic State is likely to attack areas with a higher share of Muslims and less likely to attack more Christian-occupied areas. The likelihood that the JNIM targets an area increases by three times for every additional unit to the share of Christians. Fulani militia increases the odds of targeting cells by four times with more units of the share of Christians in that cell and reduces by 99% for higher Muslim population. The odds that Boko Haram attacks a cell increases by 47 times for an additional unit to the share of Muslims in that cell. These results confirm that the distribution of religious affiliations in a cell affects the likelihood of terror attacks. Specifically, it highlights that groups like the Fulani militia, JNIM, Katiba Macina, and Ambazonian Separatists are more likely to target areas more populated by Christians. Additionally, Boko Haram, Communal Militia, and Islamic State are more likely to target Muslim-occupied areas.

4.2 Significance and relevance of the model

The Area Under the ROC Curve (AUC) is included on each table. The AUC serves as a comprehensive measure assessing the efficacy of a logistic regression model in distinguishing between positive and negative outcomes, while also gauging its predictive accuracy. In the context of modelling terror groups, the AUC ranges between 0,59 and 0,94 initially, indicating favourable performance. Notably, this performance is enhanced, reaching a range of 0,60 to 0,98, upon the inclusion of variables related to religious parity.

4.3 Hypotheses checking

After careful examination of the factors affecting the occurrence of terrorism in West Africa, we accept the hypothesis proposed in section 2 of this paper. Specifically, we confirm that the determinants of terrorist attacks differ from group to group. For example,

the odds that Boko Haram attacks a cell reduces by 0,9% for each kilometre added to cell distance from the nation border, while these odds increase by 1% for the JNIM. Also, the Fulani ethnic militia increases chances of attack for cells with higher shares of Christians, while the Islamic State reduces chances of attacks for cells with higher share of Christians.

5. Conclusion

In conclusion, this study identified the multifaceted determinants of terrorism in West Africa, shedding light on the distinct risk factors associated with each terrorist organization analysed in this research. We addressed the fundamental question of how economic, political, demographic, and religious variables influence the decision-making processes of terrorist groups when selecting their targets. Our examination has singled out seven prominent terrorist organizations operating within the region. Notably, we have discerned that the determinants of terrorism, encompassing GDP per capita, infant mortality rates, population density, democratic levels, local government control, relative wealth indices, and the proportions of Christians and Muslims, exhibit notable variations among these groups. As a result, we emphasize the imperative need to tailor counter-terrorism strategies to accommodate the unique characteristics, goals, and tactics of each terrorist organization. Such an approach is essential, given the dynamic and heterogeneous nature of terrorism in West Africa. Solutions aimed at reducing poverty, bridging the economic gap, and improving democracy and local government control might be effective against terrorism.

The data that support the findings of this study are openly available in [figshare](#).

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